Breast Ultrasound Image Classification
Using Deep Learning

Handling Multi-Class Imbalance on the BUSI Dataset

Submitted by

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Abstract

Breast cancer is one of the leading causes of death among women worldwide. Early and accurate detection significantly improves patient survival rates. This project develops a deep learning-based system to classify breast ultrasound images into three categories — *benign*, *malignant*, and *normal* — using the publicly available BUSI dataset. A pretrained ResNet50 model is employed as the backbone through transfer learning. Six strategies are systematically explored to handle the inherent class imbalance in the dataset: Baseline, Data Augmentation, Weighted Random Sampling, Focal Loss, SMOTE Hybrid, and Mixup Training. All models are trained using the PyTorch framework and evaluated on a held-out test set of 117 images using accuracy, precision, recall, and F1-score.

This report presents the methodology, experiments, results, and analysis of the BUSI breast ultrasound image classification project.

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All experiments use ResNet50 pretrained on ImageNet.
Dataset: BUSI — 780 breast ultrasound images across 3 classes.

1. Introduction

Breast cancer is one of the most common and life-threatening cancers among women worldwide. Early and accurate detection is critical for improving patient outcomes and survival rates. Ultrasound imaging is widely used in clinical settings because it is non-invasive, radiation-free, inexpensive, and accessible. However, manual interpretation of ultrasound images is time-consuming and subject to inter-observer variability among radiologists.

The rise of deep learning, particularly Convolutional Neural Networks (CNNs), has opened a promising avenue for automated and consistent medical image diagnosis. In this project, a CNN-based classifier is developed using transfer learning with ResNet50. A central challenge addressed is the significant class imbalance in the BUSI dataset, where benign cases far outnumber malignant and normal ones. Six different imbalance-handling strategies are explored and compared.

2. Problem Statement

The BUSI dataset is inherently imbalanced: benign cases account for over 55% of images, while normal cases make up only 17%. A naive classifier trained on such data tends to be biased toward the majority class, achieving high overall accuracy while performing poorly on clinically critical minority classes.

Failing to detect a malignant case (false negative) has severe consequences for patient health. The goals of this project are:

1. Develop an accurate deep learning model to classify breast ultrasound images into three categories: benign, malignant, and normal.
2. Systematically compare six class imbalance handling strategies to determine which best improves minority class sensitivity.

3. Dataset Description

The **BUSI (Breast Ultrasound Images)** dataset is a publicly available benchmark collected from 600 female patients at Badr University Hospital in 2018. It contains 780 images across three classes along with segmentation masks, which are excluded in this project.

Class	**Total**	**Proportion**	**Train (70%)**	**Test (15%)**
Benign	437	55.8%	305	66
Malignant	210	26.9%	147	31
Normal	133	17.0%	93	20
Total	**780**	**100%**	**546**	**117**

Table 1: BUSI dataset class distribution and stratified split counts.

A **stratified 70/15/15** split preserves class proportions across training, validation, and test sets. All images are converted to RGB, resized to 224×224 pixels, and normalised using ImageNet statistics (mean: 0.485, 0.456, 0.406; std: 0.229, 0.224, 0.225).

4. Methodology

The end-to-end pipeline consists of:

1. **Data Loading:** Image paths collected, mask files excluded, integer labels assigned (benign=0, malignant=1, normal=2).

2. **Splitting:** Stratified 70/15/15 split with `random_state=42`.
3. **Dataset:** Custom `UltrasoundDS` PyTorch class with on-the-fly Albumentations transforms.
4. **Training:** Each experiment trains ResNet50 for 8 epochs, saving the best checkpoint by validation loss.
5. **Evaluation:** Test set assessed with classification report and confusion matrix per experiment.
6. **Comparison:** Results aggregated into a final comparison table and bar chart.

5. Model Architecture

All experiments use **ResNet50** pretrained on ImageNet as the backbone. The final fully connected layer is replaced with a new linear layer for 3-class output:

$$\text{ResNet50 Backbone} \longrightarrow \text{Linear}(2048,\ 3) \longrightarrow \text{Softmax}$$

Training configuration: AdamW optimiser, learning rate $= 1 \times 10^{-4}$, weight decay $= 10^{-4}$, 8 epochs, batch size 16, GPU (CUDA) when available.

6. Handling Class Imbalance Techniques

6.1. Baseline

Standard CrossEntropy loss with random shuffling and no augmentation. Serves as the reference point to observe the raw effect of class imbalance.

6.2. Data Augmentation

Medical-style augmentations applied via Albumentations: horizontal flip (p=0.5), rotation $\pm 20°$ (p=0.5), brightness/contrast (p=0.4), and Gaussian blur (p=0.3). Increases training diversity without collecting new data.

6.3. Weighted Random Sampling

Each training sample is assigned weight $w_c = N/N_c$ (inverse class frequency). `WeightedRandomSampler` ensures each mini-batch contains a balanced representation of all classes.

6.4. Focal Loss

Modifies cross-entropy to down-weight easy majority examples:

$$FL(p_t) = -(1 - p_t)^{\gamma} \log(p_t), \qquad \gamma = 2$$

Hard, misclassified (typically minority) samples receive higher loss weight, focusing model learning on them.

6.5. SMOTE Hybrid

ResNet50 (frozen) extracts 2048-d feature embeddings. SMOTE synthesises new minority class samples in this feature space. A lightweight MLP (512→256 hidden units) is then trained on the balanced feature set, decoupling feature learning from classification.

6.6. Mixup Training

Pairs of training images and labels are interpolated using $\lambda \sim \text{Beta}(0.4, 0.4)$:

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j, \qquad \tilde{y} = \lambda y_i + (1 - \lambda) y_j$$

Encourages smoother decision boundaries and reduces overconfident majority class predictions.

7. Evaluation Metrics

- **Accuracy:** Overall fraction of correct predictions. Misleading on imbalanced datasets.
- **Precision (Macro):** Average per-class fraction of positive predictions that are correct.
- **Recall (Macro):** Average per-class fraction of actual positives correctly identified. *Most clinically critical* — high malignant recall means fewer missed cancers.
- **F1-Score (Macro):** Harmonic mean of precision and recall. Primary comparison metric. All macro averages weight each class equally regardless of support.

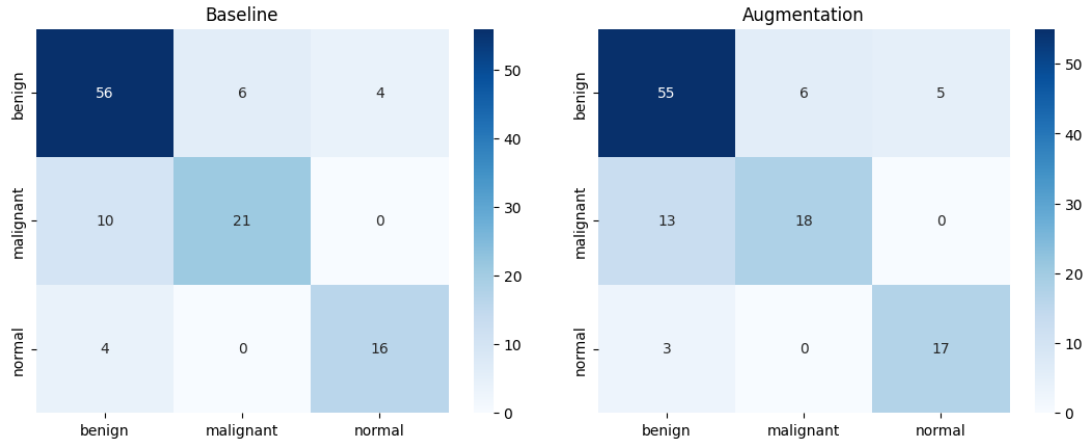
8. Results and Analysis

8.1. Per-Class Classification Results

Experiment	Class	Precision	Recall	F1	Support
Baseline	Benign	0.80	0.85	0.82	66
	Malignant	0.78	0.68	0.72	31
	Normal	0.80	0.80	0.80	20
Augmentation	Benign	0.77	0.83	0.80	66
	Malignant	0.75	0.58	0.65	31
	Normal	0.77	0.85	0.81	20
Weighted Sampling	Benign	0.82	0.80	0.81	66
	Malignant	0.69	0.77	0.73	31
	Normal	0.82	0.70	0.76	20
Focal Loss	Benign	0.82	0.77	0.80	66
	Malignant	0.73	0.77	0.75	31
	Normal	0.73	0.80	0.76	20
SMOTE Hybrid	Benign	0.85	0.80	0.83	66
	Malignant	0.67	0.77	0.72	31
	Normal	0.74	0.70	0.72	20
Mixup	Benign	0.79	0.73	0.76	66
	Malignant	0.63	0.77	0.70	31
	Normal	0.78	0.70	0.74	20

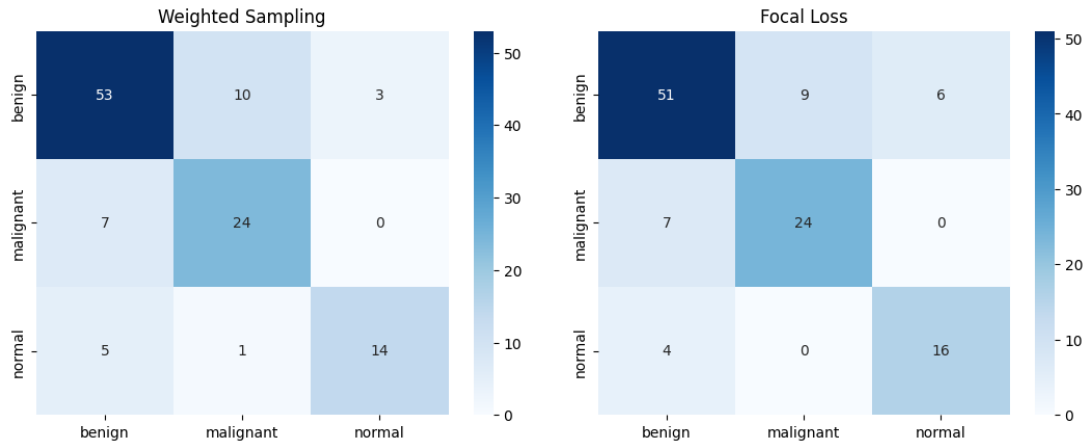
Table 2: Per-class precision, recall, and F1 for all experiments on the test set (117 images).

8.2. Confusion Matrices



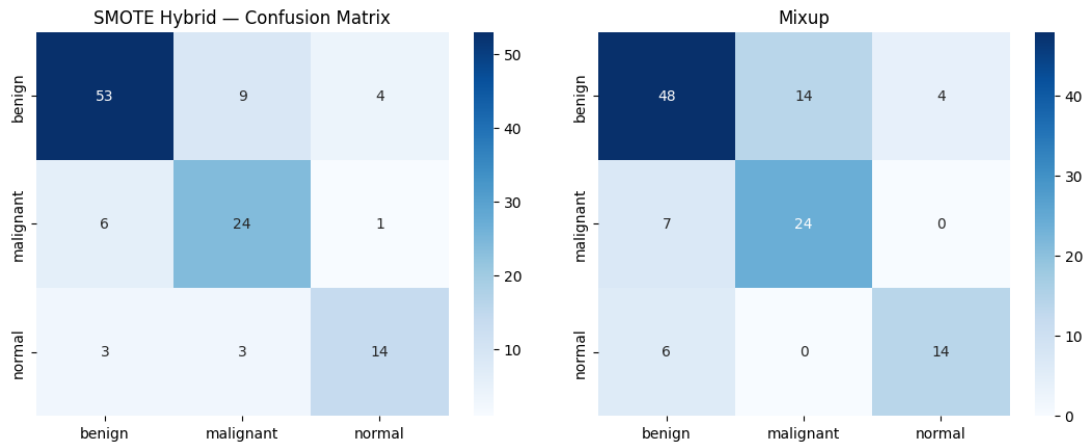
(a) Baseline

(b) Augmentation



(c) Weighted Sampling

(d) Focal Loss



(e) SMOTE Hybrid

(f) Mixup

Figure 1: Confusion matrices for all six experiments on the test set.

8.3. Overall Comparison

Experiment	Accuracy	Precision	Recall	F1 Score
Baseline	**0.7949**	**0.7926**	0.7753	**0.7826**
Augmentation	0.7692	0.7658	0.7547	0.7557
Weighted Sampling	0.7778	0.7749	0.7591	0.7644
Focal Loss	0.7778	0.7590	**0.7823**	0.7696
SMOTE Hybrid	0.7778	0.7528	0.7591	0.7542
Mixup	0.7350	0.7321	0.7338	0.7295

Table 3: Macro-averaged metrics for all experiments on the test set. Bold = best per column.

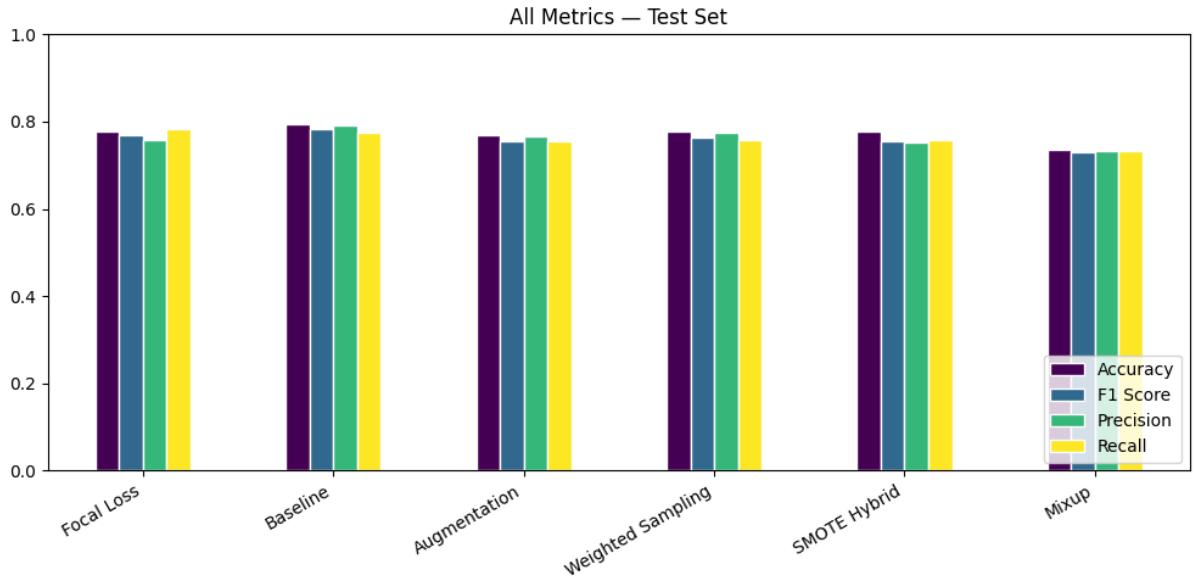


Figure 2: Comparison of all evaluation metrics across the six experiments.

9. Discussion

Why accuracy is misleading. A model always predicting "benign" achieves >56% accuracy while being clinically useless. Macro F1 and per-class recall are far more informative for imbalanced datasets.

Baseline performs strongly overall. The pretrained ResNet50 achieves the highest accuracy (0.7949) and F1 (0.7826), confirming that ImageNet features transfer well to ultrasound images. However, its malignant recall is the lowest (0.68) — it misses 32% of actual malignant cases, which is unacceptable in clinical screening.

Focal Loss best addresses clinical recall. With macro recall of 0.7823 and malignant recall of 0.77, Focal Loss achieves the most important improvement. By down-weighting easy majority examples, it directs gradient toward hard minority samples.

Weighted Sampling is equally effective for malignant recall (0.77), requiring only a dataloader-level change with no modification to the loss function — making it the simplest effective technique.

SMOTE Hybrid achieves the best benign precision (0.85) in feature space but does not surpass end-to-end models on overall metrics, as deep feature classification loses fine pixel-level

discriminability.

Augmentation and Mixup underperform. Mixup scored the lowest across all metrics (F1: 0.7295). For ultrasound images, where discriminating features are subtle textures, blending unrelated images creates ambiguous training samples that hurt generalisation.

10. Conclusion

This project systematically studied six class imbalance handling strategies for breast ultrasound classification using ResNet50 on the BUSI dataset. Key findings:

- **Baseline** achieved the best overall accuracy (0.7949) and F1 (0.7826).
- **Focal Loss** achieved the highest macro recall (0.7823) and improved malignant recall from 0.68 to 0.77 — the most clinically meaningful result.
- **Weighted Sampling** equally improved malignant recall to 0.77 with minimal implementation overhead.
- **Mixup** was the least effective technique for this dataset.
- **Accuracy alone is insufficient** — macro F1 and per-class recall must be reported for imbalanced medical datasets.

For clinical deployment, **Focal Loss combined with Weighted Sampling** is recommended as it best prioritises sensitivity on minority classes. Future work could explore ensemble methods, self-supervised medical pretraining, and class-specific threshold calibration.